Shyam Harimohan Menon

Center for Computational Astrophysics
Flatiron Institute
NY 10010, USA
smenon@flatironinstitute.org
shm-1996.github.io
shm-1996
0000-0001-5944-291X

Professional Experience

Sep 2025 – present **Flatiron Research Fellow (Joint Rutgers–Flatiron Fellow)**,
Center for Computational Astrophysics (CCA), Flatiron Institute, NY, USA

Sep 2023 – Aug 2025 **Postdoctoral Fellow (Joint Rutgers–Flatiron Fellow)**,
Dept. of Physics and Astronomy, Rutgers University, NJ, USA

Education

Oct 2019 – Sep 2023 **Doctor of Philosophy**, *Astronomy & Astrophysics*,
Research School of Astronomy & Astrophysics, Australian National University, Canberra, Australia
Thesis: The interplay of star formation and radiation feedback in the ISM.
Awarded the Charlene Heisler Prize.
Advisors: Prof. C. Federrath & Prof. M. Krumholz

Aug 2014 – June 2019 **Integrated Bachelor and Master of Technology (Hons.)**, *Engineering Physics*,
Indian Institute of Technology (BHU), Varanasi, India
Thesis: HI Fragmentation in the Turbulent Interstellar Medium.
Advisor: A/Prof. P. Dutta

Honors & Grants

July 2024 **Charlene Heisler Prize**,
Awarded by the Astronomical Society of Australia for the most outstanding PhD thesis accepted by an Australian university

Feb 2023 **Joint 4-year Rutgers-Flatiron Postdoctoral Fellowship (∼$400,000)**

Feb 2023 **CCAPP Postdoctoral Fellowship (declined)**

Feb 2023 **CITA National Fellowship (declined)**

March 2022 **RSAA Travel Grant (∼$6000)**

Oct 2021 **ANU Olin J Eggen Research Award (∼$2000)**,
Awarded to a PhD student for excellence in research

Oct 2019 **ANU International PhD Scholarship (∼$80,000)**

Oct 2019 **ANU RSAA Supplementary Research Scholarship (∼$7000)**

May 2017 **Mitacs Globalink Research Internship Award (∼$5000)**

April 2013 **Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship**

Leadership and Scientific Service

2026-2028 **Leader: International Space Science Institute (ISSI) International Team**,
Leading the competitively-selected ISSI International Team #26-690 focused on multiscale modeling of dust in the interstellar medium of galaxies in the early Universe ($z \gtrsim 6$).

Feb 2027 **Lead Organiser, Scientific Workshop on "The Lifecycle of Dense Star Clusters"**,
Lead organiser of a focused workshop consisting of ∼50 international experts on dense star cluster formation and evolution in New York City

April 2026 **Expert grant panel reviewer: National Science Foundation (NSF) Grants**,
Invited to provide expert reviews on an NSF-AAG grant panel.

July 2021 – present **Reviewer: ApJ, ApJL, MNRAS, J. Comput. Phys., JOAA, Open Journal of Astrophysics**

Feb 2022 – May 2023 **Chair, Seminars Committee, RSAA**,
Heading the seminars team; responsible for organising, hosting and inviting speakers for colloquia

Sept 2020 – Feb 2022 **Member of Seminars Committee, RSAA**,
Hosted >30 external colloquia and seminars at RSAA

Students Mentored

2 PhD students + 2 master's + 2 undergraduate students

Jan 2025 – May 2025 **Thummim Mekuria, CCA Predoctoral Project (PhD student at UC Santa Cruz)**,
Project title: The Feedback-Matter Interplay in High-z Compact Starbursts

Jan 2025 – present **Avi Chen, PhD student, Rutgers University**,
Project title: Role of Radiative Feedback in Population III Star Formation

Nov 2024 – Nov 2025 **Lisanne van Veenen, Master's student at Leiden University**,
Project title: Radiation magnetohydrodynamics simulations of Population III star formation during the epoch of reionization

March 2024 – March 2025 **Dhavala Sai Srinivas, Master's student at IISER-Kolkata**,
Project title: The escape fraction of ionising photons in magnetized turbulent media

Jan 2025– March 2025 **Frances Mathias, Undergraduate, Rutgers University**,
Project title: Stochasticity in Stellar Population Synthesis

Jan 2022 – March 2022 **Hank Hua, undergraduate student at ANU (co-supervised)**,
Project title: Effects of spiral arms on the Two Point Correlation Function

Teaching Experience

Oct 2026 **Guest Lecture, Physics 345 – Introduction to Computational Astrophysics**,
Guest lecture on high performance computing for graduate students

Sep 2025 **Guest Lecture, Physics 610 – Interstellar Matter**,
Guest lectures on ISM physics for graduate students

Nov 2024 **Guest Lecture, Physics 345 – Introduction to Computational Astrophysics**,
Guest lecture on parallel computing for graduate students

Oct 2022 **Guest Lecture, ASTR 4012 – Astrophysical Gas Dynamics**,
Guest Lecture on The Physics of Radiation Hydrodynamics for graduate students

Feb 2022 – May 2023 **Teaching Assistant, ASTR 1003 – Astronomy & Space**,
Respond to undergraduate student queries on the course forum and assist in tutorials

Outreach Activities

Sep 2026 **Public talk, *Mortal Matters, Greenwood Public Cemetery***,
Delivered an invited public talk on "All Ends are Beginnings" at the "Mortal Matters" event held at the Greenwood Public Cemetery.

Aug 2026 **Public talk, *Astronomy on Tap (AoT), Bengaluru***,
Delivered an invited AoT public outreach talk on "Survival of the Fittest: Lessons from Ancient Stellar Nurseries" that obtained press coverage.

Sep 2024 – May 2025 **STEM professional volunteer: Letters to a Pre-Scientist**,
Program to exchange letters with middle school students to broaden exposure to the day-to-day of STEM researchers.

July 2021 – Dec 2022 **Hosting public nights, Mount Stromlo Observatory**,
Hosting telescope viewing nights at the observatory for public school students and adults as part of the outreach team at ANU.

Invited and Prize Talks

April 2027 **Deepdish-Baguette: Stars, Black Holes, and Feedback From Cosmic Dawn to Cosmic Noon**

Nov 2026 **Department of Physics Colloquium: University of Texas, Dallas**

Sep 2026 **Colloquium: Tata Institute of Fundamental Research, Mumbai**

Sep 2026 **Seminar: Raman Research Institute (RRI), Bengaluru**

May 2026 **Astrophysics Seminar: American Museum of Natural History**

April 2026 **Star Formation/ISM Rendezvous: Princeton University**

Jan 2026 **The growth of galaxies in the early Universe XI, Sesto, Italy**

Dec 2025 **Theoretical Astrophysics Seminar, UC Berkeley**

Aug 2025 **Santa Cruz Galaxy Workshop, UC Santa Cruz**

July 2025 **Review Talk, Numerical Recipes in Star Formation, Lorentz Center, Leiden University**

Aug 2024 **Cosmic Dawn Revealed by JWST, Kavli Institute of Theoretical Physics**

June 2024 **Charlene Heisler Prize Talk, ASA Annual Meeting**

March 2024 **ISM Salon, CCA Flatiron Institute**

March 2024 **Radiation Transport in Astrophysics, Heidelberg**

July 2023 **Seminar: Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune**

June 2022 **Seminar: Centre de Recherche Astrophysique de Lyon (CRAL)**

Dec. 2021 **Colloquium: Indian Institute of Astrophysics, Bangalore**

March 2020 **Modelling High-Mass Stellar Feedback Workshop, Tübingen, Germany**

Contributed Talks (*selected*)

July 2026 **Cake Talk: Cosmic Dawn Center, Copenhagen**

June 2026 **EAS Annual Meeting (Special Session 5 and 7), Lausanne, Switzerland**

March 2026 **Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration, Tokyo, Japan**

April 2025	**Escape of Lyman radiation from galactic labyrinths, Crete, Greece**
April 2024	**STScI Spring Symposium, Baltimore**
Feb. 2024	**Building Galaxies From Scratch, Vienna**
Nov. 2022	**CIERA Theory Seminar, Northwestern University**
Nov. 2022	**CCAPP Seminar, Ohio State University**
Oct. 2022	**Coffee Seminar, Institute for Advanced Study**
Oct. 2022	**Cosmology Seminar, MPA Garching**
Aug. 2022	**Star Formation in Different Environments, Quy Nhon, Vietnam**
July 2022	**A Holistic View of Stellar Feedback and Galaxy Evolution, Ascona**
July 2022	**Planet and Star Formation (PSF) Seminar - MPIA, Heidelberg**
June 2022	**NBIA Workshop on Radiation Transfer in Astrophysics, Copenhagen**
Nov. 2021	**IAU Symposium 362 - The Predictive Power of Computational Astrophysics**

List of Publications

33 publications: 10 first-author, 5 second-author; h-index: 15; total citations: 628 (ADS listing)

First-author publications

1. Non-equilibrium chemistry and radiation hydrodynamics in metal-poor and metal-rich interstellar gas
   Menon, S., Sharda, P., Burkhart, B., Grassi, T. ApJ subm., 2026, arXiv:2610.17577
2. Lyman-alpha Radiation Pressure in Dense Star Clusters: Implications for Star Formation and Winds at Cosmic Dawn
   Menon, S., Smith, A. ApJ accepted (in press), 2026, arXiv:2605.13982.
3. Bursts of Star Formation and Radiation-driven Outflows Produce Efficient LyC Leakage from Dense Compact Star Clusters
   Menon, S., Burkhart, B., Somerville, R., Thompson, T., Sternberg, A. ApJ, 2025, 987, 12
4. The Interplay between the Initial Mass Function and Star Formation Efficiency through Radiative Feedback at High Stellar Surface Densities
   Menon, S., Lancaster, L., Burkhart, B., Somerville, R., Dekel, A., Krumholz, M. ApJL, 2024, 967, L28
5. Outflows driven by direct and reprocessed radiation pressure in massive star clusters
   Menon, S., Federrath, C., Krumholz, M. MNRAS, 2023, 521, 5160
6. Infrared radiation feedback does not regulate star cluster formation
   Menon, S., Federrath, C., Krumholz, M. MNRAS, 2022, 517, 1313
7. VETTAM: a scheme for radiation hydrodynamics with adaptive mesh refinement using the variable Eddington tensor method
   Menon, S., Federrath, C., Krumholz, M., Kuiper, R., Wibking, B., Jung, M. MNRAS, 2022, 512, 401
8. The dependence of the hierarchical distribution of star clusters on galactic environment
   Menon, S., Grasha, K., Elmegreen, B., Federrath, C., Krumholz, M., Calzetti, D., et al. MNRAS, 2021, 507, 5542
9. On the compressive nature of turbulence driven by ionizing feedback in the pillars of the Carina Nebula
   Menon, S., Federrath, C., Klaassen, P., Kuiper, R., Reiter, M. MNRAS, 2021, 500, 1721
10. On the turbulence driving mode of expanding H II regions
    Menon, S., Federrath, C., Kuiper, R. MNRAS, 2020, 493, 4643

Co-author publications with significant contributions

1. Radiative Feedback in Population III Protostellar Growth: The Impact of HI Shielding
   Chen, A., **Menon, S.**, Burkhart, B., Sharda, P., Williams, C., Naoz, S., et al. arXiv e-prints, 2026, arXiv:2604.00197
2. Planets and planetesimals at cosmic dawn: vortices as planetary nurseries
   Eriksson, L., **Menon, S.**, Carrera, D., Lyra, W., Burkhart, B. MNRAS, 2025, 542, 641
3. Magnetic fields limit the mass of Population III stars even before the onset of protostellar radiation feedback
   Sharda, P., **Menon, S.**, Gerasimov, R., Bromm, V., Burkhart, B., Haemmerlé, L., et al. MNRAS, 2025, 541, L1
4. Population III star formation in the presence of turbulence, magnetic fields, and ionizing radiation feedback
   Sharda, P., **Menon, S.** MNRAS, 2025, 540, 1745
5. First extragalactic measurement of the turbulence driving parameter: ALMA observations of the star-forming region N159E in the Large Magellanic Cloud
   Sharda, P., **Menon, S.**, Federrath, C., Krumholz, M., Beattie, J., Jameson, K., et al. MNRAS, 2022, 509, 2180
6. Blue Monsters and Dusty Descendants: Reconciling UV and IR Emission from Galaxies from z∼7, up to z = 14
   Sommovigo, L., Lancaster, L., **Menon, S.**, O'Leary, J., Somerville, R., Bryan, G. ApJ, 2026, 1003, 170
7. Density-modulated star formation efficiency: implications for the observed abundance of ultraviolet luminous galaxies at $z > 10$
   Somerville, R., Yung, L., Lancaster, L., **Menon, S.**, Sommovigo, L., Finkelstein, S. MNRAS, 2025, 544, 3774
8. Radiation magnetohydrodynamics simulations of Population III star formation during the epoch of reionization
   van Veenen, L., Sharda, P., Viti, S., **Menon, S.** A&A, 2026, 708, A61
9. The Coevolution of Stellar Wind-blown Bubbles and Photoionized Gas. I. Physical Principles and a Semianalytic Model
   Lancaster, L., Kim, J., Bryan, G., **Menon, S.**, Ostriker, E., Kim, C. ApJ, 2025, 989, 42
10. Investigating the Young Stellar Populations and Hierarchies in Nearby Galaxies with the UVIT. III. Evidence for a Largest Scale of Correlated Stellar Structures and a Non-universal Fractal Dimension
    Shashank, G., Subramanian, S., **Menon, S.**, Mondal, C., Grasha, K., de Grijs, R., et al. arXiv e-prints, 2026, arXiv:2609.01065
11. Linking Stellar Populations to H II Regions across Nearby Galaxies. II. Infrared Reprocessed and UV Direct Radiation Pressure in H II Regions
    Pathak, D., Leroy, A., Thompson, T., Lopez, L., Barnes, A., Dale, D., **incl. Menon, S.**, et al. ApJ, 2025, 982, 140

Contributions to collaborative efforts

1. Wandering intermediate-mass black holes and off-nuclear tidal disruption events from early star clusters
   Williams, C., Naoz, S., Zhou, Y., Akiba-Lewis, T., Vogelsberger, M., Burkhart, B., **incl. Menon, S.**, et al. arXiv e-prints, 2026, arXiv:2610.20719
2. The THESAN-ZOOM project: the delayed rise of Lyman continuum escape fractions in bursty high-redshift galaxies
   Mehrotra, P., Smith, A., Bian, Y., McClymont, W., Puchwein, E., Kannan, R., **incl. Menon, S.**, et al. arXiv e-prints, 2026, arXiv:2610.30265

3. Investigating the young stellar populations and hierarchies in nearby galaxies with the UVIT. II. Presenting the properties of 25,000 UV-detected star-forming clumps
   Shashank, G., Subramanian, S., Mondal, C., **Menon, S.**, Subramaniam, A. arXiv e-prints, 2026, arXiv:2606.12254
4. Tracing hierarchical star formation out to kiloparsec scales in nearby spiral galaxies with UVIT
   Shashank, G., Subramanian, S., Muraleedharan, S., **Menon, S.**, Mondal, C., Krishna, S., et al. A&A, 2025, 693, A188
5. An FUV-optical Approach for Studying Hierarchical Star Formation in Nearby Galaxies with UVIT
   Ananthu, S., Shashank, G., Subramanian, S., Jayanth, R., **Menon, S.**, Mondal, C., et al. ApJ, 2026, 1002, 220
6. A cosmological framework for stellar collisions at high redshift in proto-globular clusters, nuclear star clusters, and Little Red Dots
   Williams, C., Naoz, S., Rose, S., Burkhart, B., Yoshida, N., Chen, A., **incl. Menon, S.**, et al. arXiv e-prints, 2026, arXiv:2603.26872
7. Observing Compact Population III Star Clusters and the Presence of Cosmic Streaming
   Williams, C., Naoz, S., Yoshida, N., Lake, W., Burkhart, B., Marinacci, F., **incl. Menon, S.**, et al. ApJ, 2026, 997, 202
8. Supersonic Turbulence in Primordial Halos: a Comparison with and without the Stream Velocity
   Chen, A., Lake, W., Williams, C., Burkhart, B., Naoz, S., **Menon, S.**, et al. ApJ, 2025, 995, 65
9. Detecting Molecular Hydrogen (H_2) Emission at Cosmic Dawn
   Johnson, M., Burkhart, B., D'Eugenio, F., Tacchella, S., Maiolino, R., Bialy, S., **incl. Menon, S.**, et al. ApJ, 2025, 992, 196
10. ΛCDM Star Clusters at Cosmic Dawn: Stellar Densities, Environment, and Equilibrium
    Williams, C., Naoz, S., Lake, W., Burkhart, B., Marinacci, F., Vogelsberger, M., **incl. Menon, S.**, et al. ApJ, 2025, 990, 135
11. A 260 pc resolution ALMA map of HCN(1→0) in the galaxy NGC 4321
    Neumann, L., Bigiel, F., Barnes, A., Gallagher, M., Leroy, A., Usero, A., **incl. Menon, S.**, et al. A&A, 2024, 691, A121
12. Hidden Gems on a Ring: Infant Massive Clusters and Their Formation Timeline Unveiled by ALMA, HST, and JWST in NGC 3351
    Sun, J., He, H., Batschkun, K., Levy, R., Emig, K., Rodríguez, M., **incl. Menon, S.**, et al. ApJ, 2024, 967, 133

Popular science articles

1. How stars shape galaxies
   Orr, M., **Menon, S.**, Burkhart, B. Physics Today, 2025, Issue 5, id. 44279